Report on Domain Specific Training – Augmented Reality App Development

Academic Year 2025–26

The Department of Computer Science and Design is committed to preparing students not only for academic excellence but also for successful careers in the ever-evolving field of engineering. To bridge the gap between academic learning and industry requirements, the department has introduced structured career paths supported by domain-specific training, certification programs, and self-learning opportunities. These initiatives enable students to gain specialized knowledge, practical exposure, and industry-relevant skills in emerging technologies.

Engineering offers numerous specialized domains, each requiring unique technical expertise. Therefore, the department has carefully identified multiple career paths and provides focused domain-specific training from the fourth semester to the sixth semester. Students are encouraged to select a specialization based on their interests, aptitude, and career aspirations. In addition to technical education, emphasis is also placed on communication skills, aptitude, soft skills, personality development, and pre-placement training, ensuring students become competent and industry-ready professionals.

As part of the Domain Specific Training (DST) initiative for the academic year 2025–26, the department organized a comprehensive training program on “Fundamentals for XR Development” followed by “Augmented Reality (AR) App Development” for 3rd year students in association with ED Spire Research Centre. **EDspire Research** is an educational technology and research organization based in Mysore, India, specializing in VR, AR, AI, and digital pedagogy. Founded by Dr. Thotreingam Kasar, the center provides skill enhancement programs, internships, and curriculum development for institutions. The training was conducted at ATME College of Engineering, Mysuru Campus, from 19 September 2025 to 14 November 2025, with the objective of equipping students with practical knowledge and hands-on experience in Extended Reality (XR) technologies.

The training focused on introducing students to modern AR technologies and enabling them to develop real-world mobile applications using industry-standard development tools. Students learned the complete AR application development lifecycle, beginning with the fundamentals of Augmented Reality and progressing toward advanced topics such as image tracking, face tracking, gesture recognition.

Throughout the training program, students actively participated in practical laboratory sessions, live demonstrations, project development activities, and collaborative learning exercises. They developed several AR applications, including Image Target Applications, ID Card Applications, Living Portrait Applications, Face Filter Applications, AR Placement Applications, Maze Games, and Portal Cube Applications. These projects significantly enhanced students' programming abilities, creativity, problem-solving skills, and understanding of real-world AR implementation.

Topics Covered During the Domain Specific Training

- Introduction to Augmented Reality, Vuforia Setup, and Pre-Assessment.
- Building Vuforia Dataset and Image Targets.
- Development of an ID Card Augmented Reality Application.
- Building a Living Portrait Application.
- Introduction to AR Foundation, Plane Detection, and AR Raycasting.
- Implementation of AR Gestures and development of an AR Place Application.
- Face Tracking and development of a Face Filter Application.
- Image Tracking and development of a Maze Game.
- Working with GPS Sensor Data and Accelerometer-Based AR Applications.
- Play Store Deployment

To measure the effectiveness of the training, a Pre-Assessment was conducted before the commencement of the sessions, followed by a Post-Assessment at the end of the program. A Practical Laboratory Assessment was also conducted, where each student successfully developed and deployed an Augmented Reality application on their mobile device. The assessment results indicated a remarkable improvement in students' understanding of AR concepts, with all participants showing better performance in the post-assessment compared to the pre-assessment.

Feedback collected from students revealed high levels of satisfaction regarding the quality of the training, hands-on activities, practical exposure, trainer effectiveness, and achievement of learning outcomes. Most students expressed increased confidence in developing Augmented Reality applications independently after completing the program.

Conclusion

The Domain Specific Training on Augmented Reality App Development proved to be a highly successful initiative in enhancing the technical competency of Computer Science and Design students. Through intensive hands-on training, industry interaction, project-based learning, assessments, and application deployment, students acquired valuable practical experience in one of today's most promising technologies.

The combination of career path identification and domain-specific training provides students with a strong foundation for professional success. These initiatives not only prepare students for campus placements and higher education but also encourage lifelong learning and innovation. The department continues to strengthen industry-academia collaboration to ensure that graduates are technically proficient, professionally confident, and well-equipped to meet the evolving demands of the global engineering industry.


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